
CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Interim Progress Review

**[Reinforcement Learning to improve AI
behaviour and adaptability in a game
environment]**

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Course: Computer Games Technology

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Reporting Period: [3/11/25] to [12/12/25]

1.0 Log of Completed Activities & Milestones

Since submitting the Interim Project Report, I have focused on the environment setup, tool configuration, ML-Agents integration and deciding on what the final game will be.

Key tasks completed during this period include:

Project Setup and Tools

- Created a GitHub repository and project Kanban aligned with the Gantt schedule.
- Installed the important AI related unity packages (ML-Agents, AI Navigation, AI Inference and the Unity Test Framework).

Game Concept and Environment Development

- Made six concept ideas and selected Predator and Prey Arena as the final project concept based on RL suitability and supervisor feedback.
- Created a prototype arena and imported environment assets (ALP and Nobiax/Yughues).
- Added temporary capsule bodies for both the Predator agent and player.

Player and ML Agent Implementation

- Set up input system and created basic first-person movement.
- Added predator agent script with raycasts and the behaviour parameters.
- Built an initial reward system and a YAML training configuration.

Sensory and Debug Systems

- Added vision cone, LOS logic, hearing system and a remodelled multipoint trail system.
- Made an on-screen debug UI showing the agents state, rewards, LOS and sensory debugging.

Training Progress

- Completed the first 1.5 training run.
- Updated rewards, observation and YAML file to stable training.
- Predator agent can now achieve positive mean reward and shows basic tracking.

Work In Progress

- The player model is mostly complete, but the enemy model, animations and abilities are incomplete.
- Updating the simplified arena continuously for better training.

Supervisor Meetings and Engagement

- 31/10/25 – Literature Review Feedback: Required the expansion of RL topics, additional sources and clearer research gaps.
- 7/11/25 – Planning Feedback: Avoid only focusing on RL, improve comparisons and document work more thoroughly.

- 14/11/25 – Game Ideas: Estimated the game concepts feasibility and chose the best choice.
- 21/11/25 – Concept Confirmation: Confirmed Predator and Prey arena as the main project, while Adaptive Maze Runner as the sub-goal.
- 5/12/25 – Pre-Interim Review Feedback: Strengthen reflection, add kanban evidence, update Gantt chart and ReadMe.

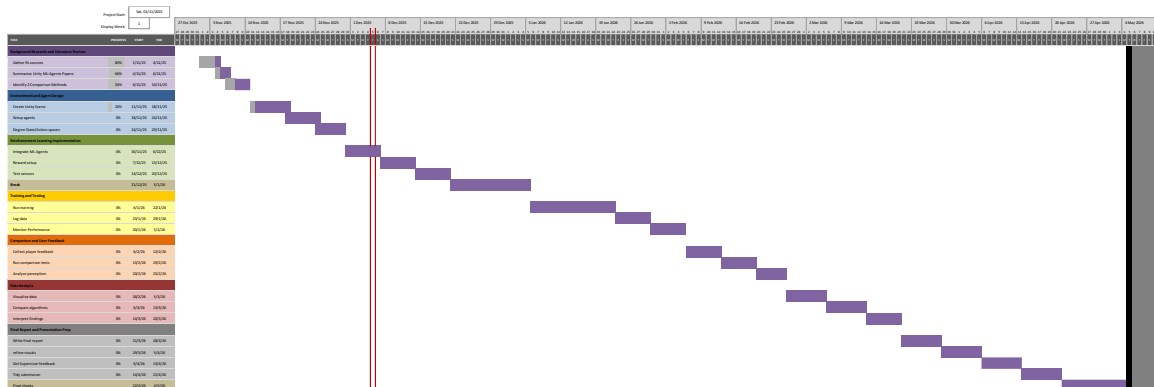
These meetings occurred weekly and were used as reflection checkpoints, which helped guide improvements to timeline expectations, concept scope and documentation practices.

2.0 Reflection and Iteration

Reflection occurs weekly through supervisor feedback, milestone reviews and the Gantt chart. The adjustments mainly came from fixing conflicts in the ML agent and python/ONNX, which caused a delay in training by 10 days. Those helped improve my workflow, as I now check my environments earlier, use smaller test scenes and update the reward structures continuously based on the training log.

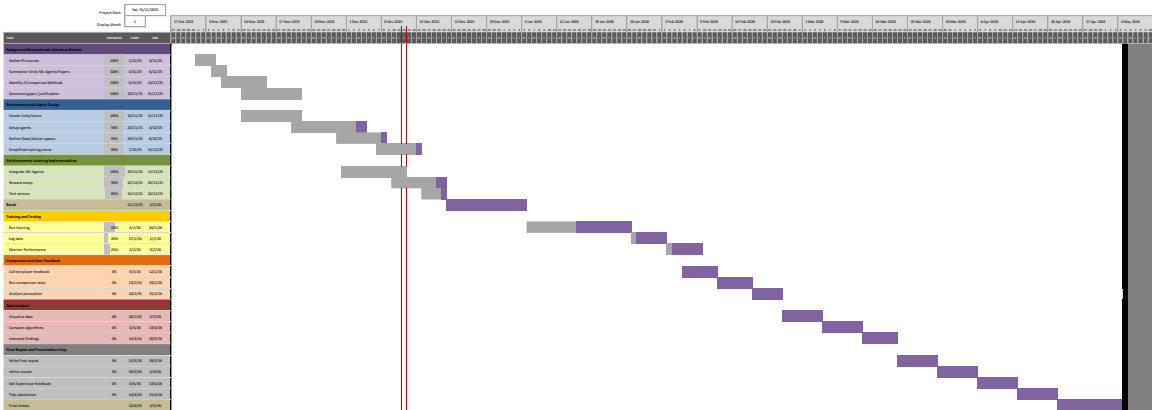
2.1 Progress Against Original Timeline

Figure 1: Previous Gantt Chart



[View full Gantt Chart here](#)

Figure 2: Updated Gantt Chart



[View full Gantt Chart here - Updated](#)

Figure 3: Kanban GitHub

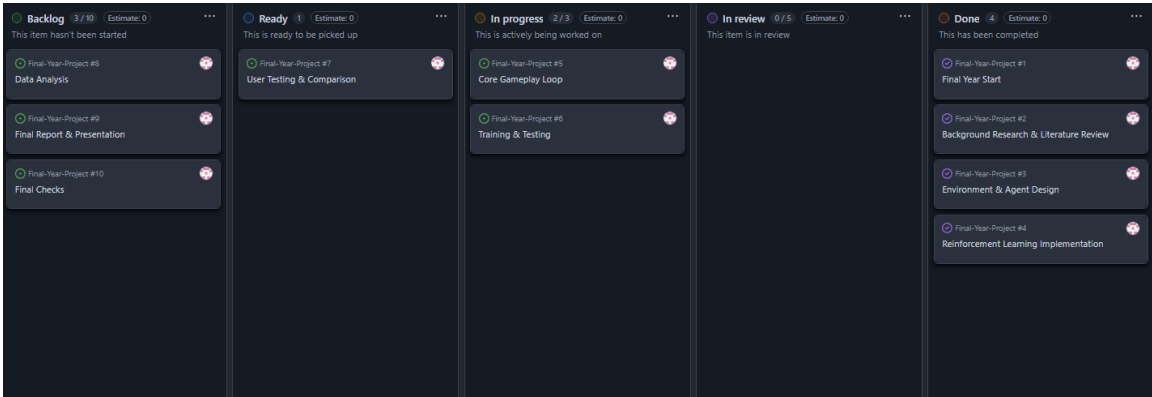
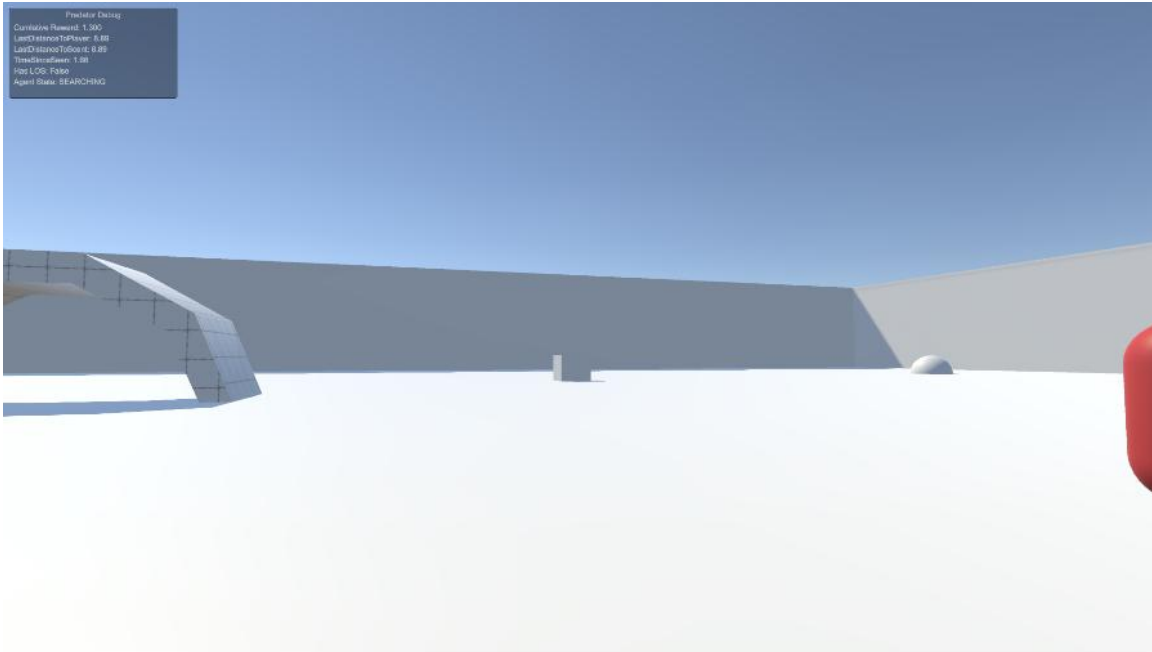


Figure 4: image of the Game Demo



[Link to video](#)

This image with the video link shows the predator debug UI being show, like its state and rewards its getting.

Current Project Status: Slightly behind Schedule

The updated Gantt chart shows the significant progress across environment design, sensory systems and early training; however, the RL integration took longer than expected. The Kanban proves that 4 out of the 10 major task have been completed with 2 major task being in progress. Furthermore, the project is slightly behind the schedule because of the conflicts the python and ONNX caused, which delayed the training for the ML agent by around 10 days.

Overall, the timeline remains achievable through minor adjustments. Which can be achieved easily through:

- Using a simplified training map.
- Decreasing December break and Increasing January training/testing time.
- Reducing scope, like making the Adaptive Maze Runner sub-goal optional.

2.2 Challenges Encountered & Resolutions

Table 1: represent the challenge encountered and resolutions

Challenge Encountered	Impact on Project	Action Taken / Planned Solution
Package conflicts between ML-Agents, ONNX, Python.	Delayed a week of training.	Downgraded Python to 3.9, installed compatible PyTorch, protobuf, onnxruntime and rebuilt the environment.
Duplicate Behaviour Parameters.	Training episodes ended instantly.	Removed extra behaviour component from gameobject called GameAcademy.
Input System not functioning.	Delayed movement and ML testing.	Fixed by wiring action events correctly and swapping mouse input mode to delta.
ONNX model export failing.	Interrupted training runs.	Installed compatible ONNX Versions after resolving NumPy conflicts.
Large training arena.	Slowed early training progress.	Created a simplified minimal training environment.
Observation size mismatch	Agent can't learn.	Updated space size to match observations in agent script.
Incorrect .bat file path.	Prevented training from starting.	Updated batch file path.
Unity agent script conflict.	Stopped agent from moving.	Removed the script.

3.0 Action Plan: Goals for Next Period

Key objectives to be met by 2/3/26:

1. Complete the ML agent training phase.
2. Implement a Behaviour tree version of the Predator for comparison.
3. Collect Early player Feedback for perception testing.
4. Begin Comparison testing between RL and Behaviour tree.
5. Prepare and organise data for the analysis phase.

4.0 Other Considerations

Due to time lost on ML-Agents configuration, the Adaptive Maze Runner sub-goal will likely be removed unless the Predator and Prey arena reaches stability ahead of schedule. The core project remains the same, evaluating RL-driven predator behaviour in a game environment.

5.0 Questions for Supervisor

1. **Question:** my refined Predator and Prey concept scoped appropriately?
Answer: Yes, but the evaluation must compare RL with an alternate method like behaviour tree or NavMesh, as comparison gives meaning to evaluating the RL performance.
2. **Question:** Should I prioritise sensory complexity (vision and hearing) or focus on perfecting one first?
Answer: Go for the common sensory types used for RL agents and add other sensors when time is available.
3. **Question:** For comparison against scripted Ai, is a Behaviour Tree sufficient as a baseline?
Answer: Yes, behaviour trees are appropriate for comparison and tie directly to question 1. Both game versions should use the same environment.
4. **Question:** Do you recommend simplifying the arena to accelerate training performance until the intermediate training phase is stable?
Answer: Don't oversimplify the scene, instead find a balanced difficulty by playtesting the environment manually to ensure it challenges the agent without confusing it.